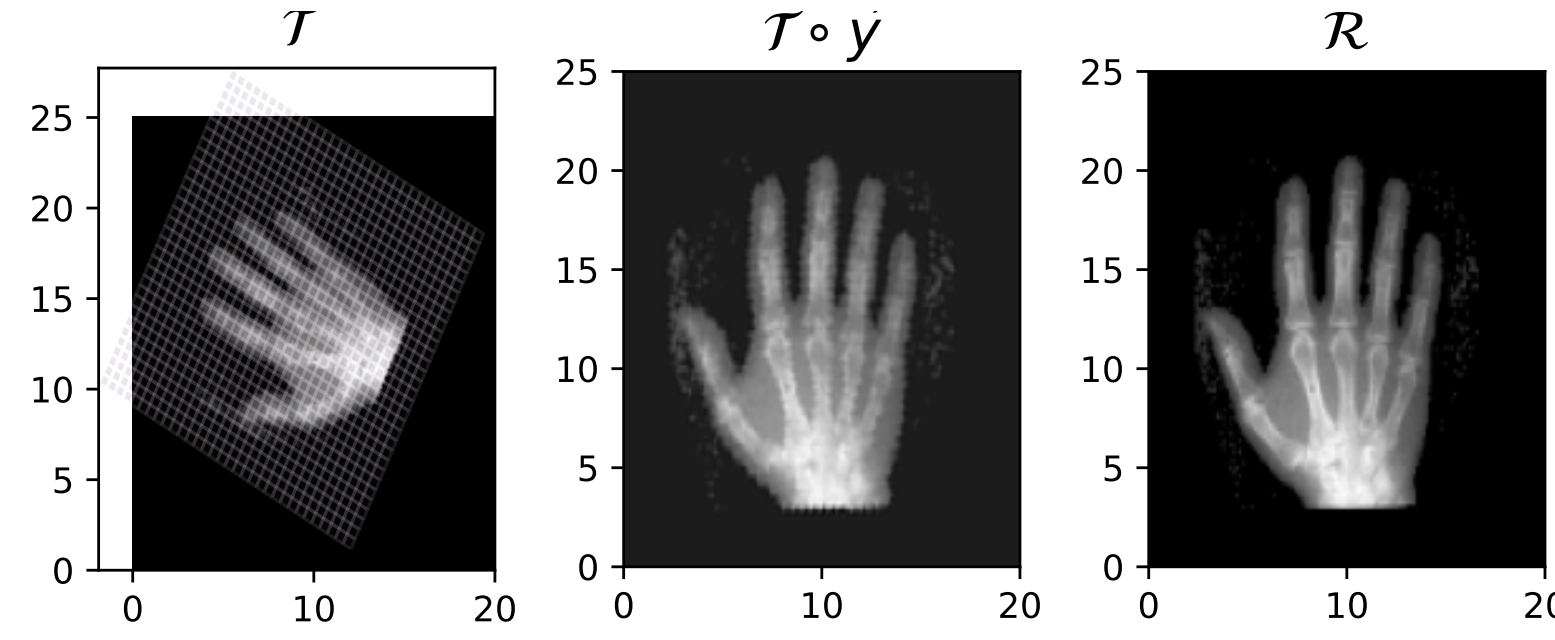


## Image Registration



We consider two images  $\mathcal{T}$  and  $\mathcal{R}$  represented as real-valued functions of a spatial domain  $\Omega \subset \mathbb{R}^2$ , and seek to find a transformation  $\tilde{y}(\cdot)$  on  $\Omega$  such that  $\mathcal{T} \circ \tilde{y} = \mathcal{R}$ . We approach this problem with a discrete optimization framework, where we minimize the loss function

$$\ell(\mathbf{w}) = \frac{1}{2} \|\mathbf{S}(\mathbf{w})\mathbf{d} - \mathbf{f}\|^2 + \lambda \text{Reg}(\mathbf{w}),$$

where  $\mathbf{d}$  and  $\mathbf{f}$  are respective vectors of  $n$  evaluations of  $\mathcal{T}$  and  $\mathcal{R}$  on the same regular grid  $\tilde{x}_1, \dots, \tilde{x}_n$ ,  $\mathbf{w}$  is a vector of  $p$  parameters determining the transformation  $\tilde{y}(\cdot; \mathbf{w})$ ,  $\mathbf{S}(\mathbf{w})$  is the matrix evaluating  $\mathcal{T}(\tilde{y}(\tilde{x}_i; \mathbf{w}))$  using a spline interpolant of the values in  $\mathbf{d}$ , and  $\lambda \text{Reg}(\mathbf{w})$  is a regularization term enforcing "feasibility" of  $\tilde{y}(\cdot; \mathbf{w})$ .

## Automatic Differentiation

Packages such as Jax and Pytorch provide *automatic differentiation* capabilities which compute derivatives of complex mathematical functions by using the chain rule to simplify expressions.

### Example: Hessian Computation

Without automatic differentiation, computing the Hessian of the Image Registration loss would require complex analytic computations. Using PyTorch, a single function call

```
torch.func.hessian(lossfn, argnums=0)(params)
```

can compute the Hessian  $\nabla^2 \ell$  of the loss.

## Neural ODEs

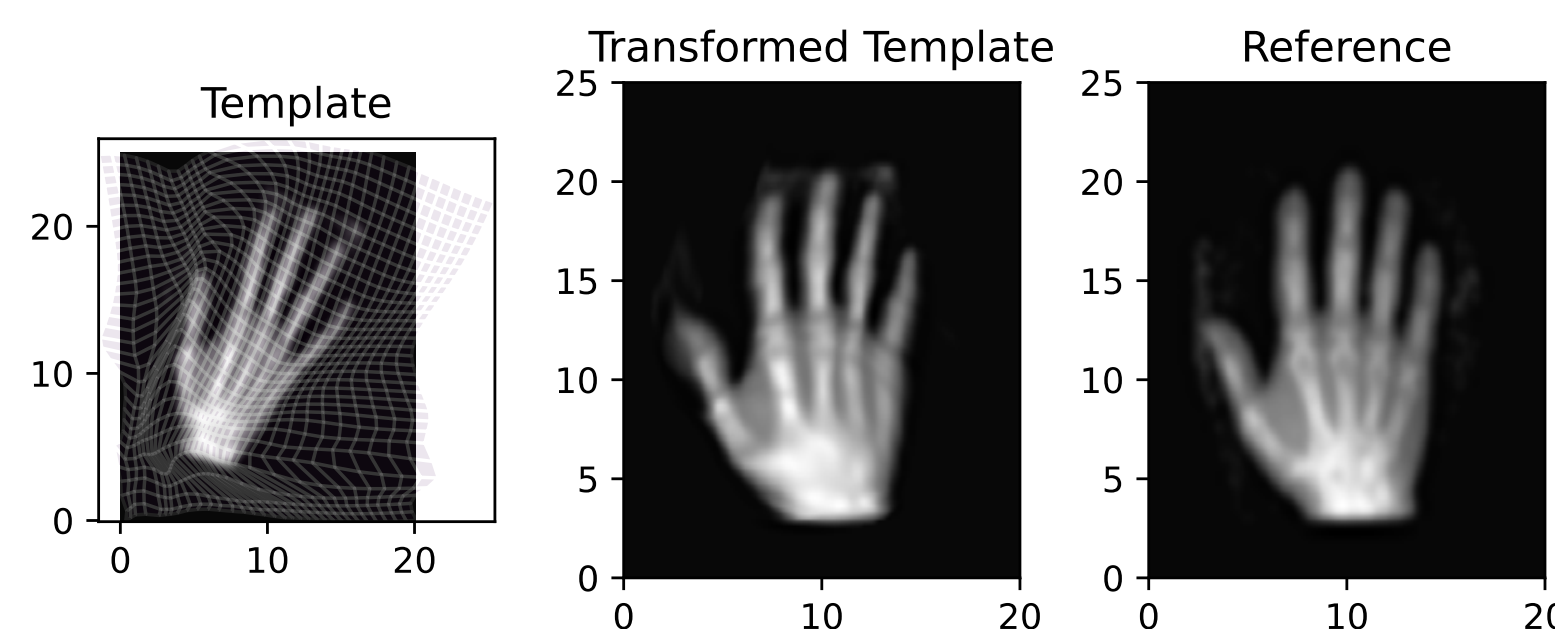
A Neural ODE [1] is an initial value problem of the form

$$\frac{d\tilde{z}_{\tilde{x}}}{dt}(t) = \tilde{v}(\tilde{z}_{\tilde{x}}(t), t; \mathbf{w}), \quad \tilde{z}_{\tilde{x}}(0) = \tilde{x},$$

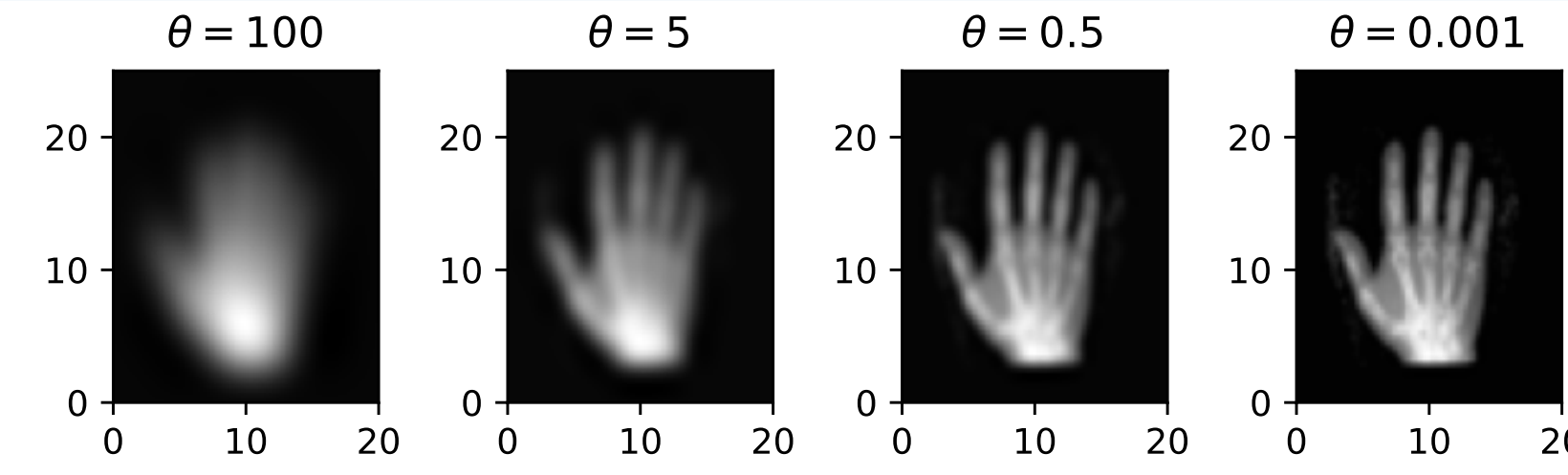
where the velocity function  $\tilde{v}(\cdot; \cdot)$  is given by a neural network. The solution gives a continuous-time function

$$\tilde{y}(\tilde{x}, t; \mathbf{w}) = \tilde{z}_{\tilde{x}}(t) = \tilde{x} + \int_0^t \tilde{v}(\tilde{z}_{\tilde{x}}(t'), t'; \mathbf{w}) dt'$$

which can be evaluated using a numerical method and trained by backpropagation through the solver. This approach has the benefit of giving diffeomorphic transformations with low dimensional parameterizations.



## Homotopy Optimization



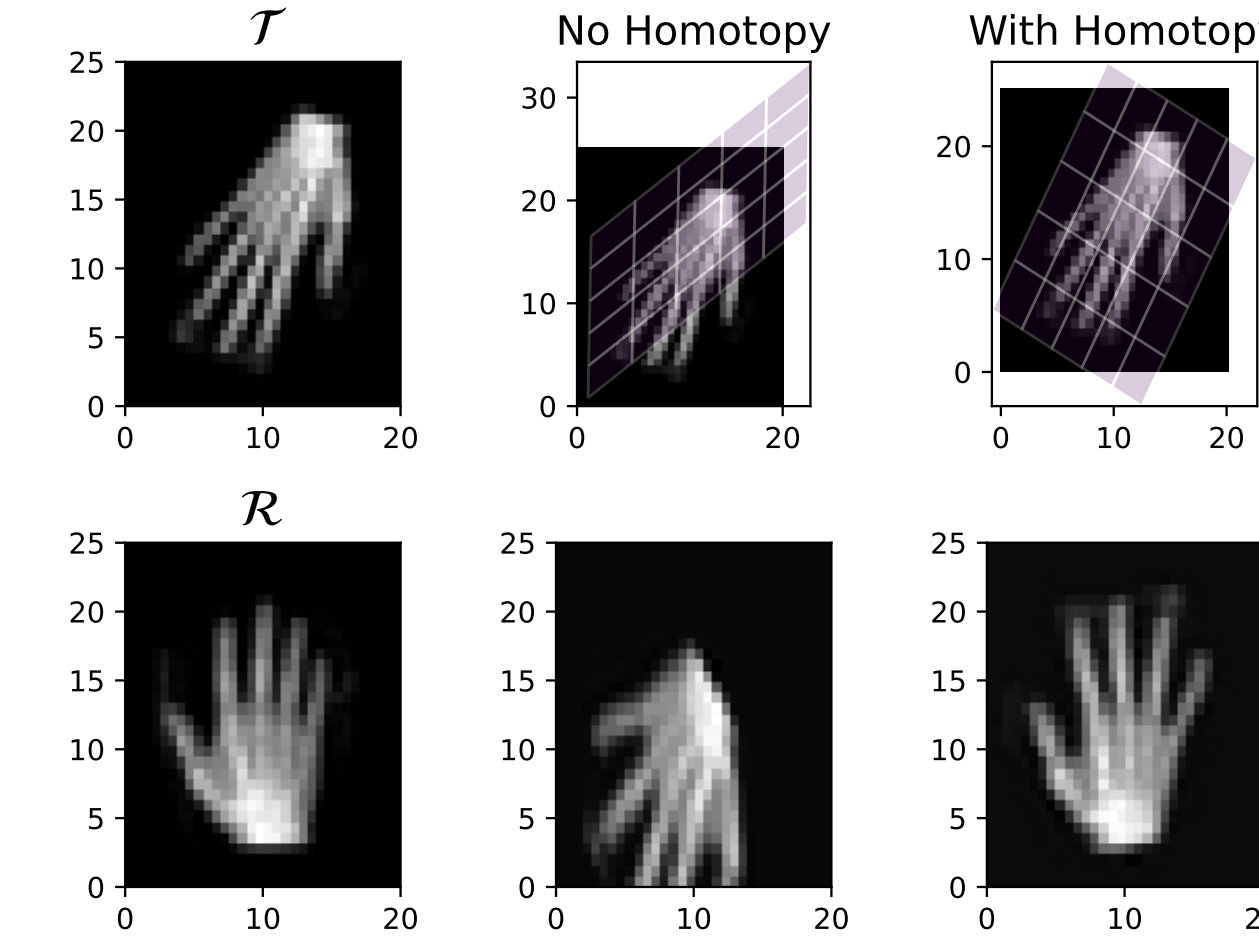
Consider a minimizer  $\mathbf{w}^*$  for a high  $\theta$  registration problem, defined by:

$$\mathbf{w}^*(\theta) = \arg \min \ell(\mathbf{w}^*(\theta), \theta).$$

By the optimality condition, the relationship between  $\mathbf{w}^*$  and  $\theta$  is governed by the following ordinary differential equation (ODE):

$$\frac{d\mathbf{w}^*}{d\theta} = -(\nabla_{\mathbf{w}}^2 \ell(\mathbf{w}^*(\theta), \theta))^{-1} \frac{d}{d\theta} \nabla_{\mathbf{w}} \ell(\mathbf{w}^*(\theta), \theta). \quad (1)$$

This approach enables tracing the minimizer from a smooth to a non-smooth registration problem.



## Super-Resolution

We reconstruct an unknown high resolution image  $\mathbf{f}$  from a given sequence of low resolution images  $\mathbf{d}^{(j)} = \mathbf{d}$  that are each a transformed, down-sampled version of  $\mathbf{f}$ . With a given down-sampling matrix  $\mathbf{K}$ , we find transformations  $\mathbf{S}^{(j)}$  parameterized by  $\mathbf{w}$  so that we minimize the loss given by

$$\ell(\mathbf{w}, \mathbf{f}^{(0)}) = \frac{1}{2} \sum_{j=1}^n \|\mathbf{K}\mathbf{S}^{(j)}(\mathbf{w})\mathbf{f}^{(0)} - \mathbf{d}^{(j)}\|^2 + \frac{\lambda}{2} \|\mathbf{L}\mathbf{f}^{(0)}\|^2 \quad (2)$$

where  $\frac{\lambda}{2} \|\mathbf{L}\mathbf{f}^{(0)}\|^2$  is the regularization term with some linear operator  $\mathbf{L}$  [2].

For a fixed  $\mathbf{w}$ ,  $\mathbf{f}^{(0)}(\mathbf{w})$  is the solution to a linear least squares problem. We can optimize the variable projected loss  $\ell(\mathbf{w}) = \ell(\mathbf{w}, \mathbf{f}^{(0)}(\mathbf{w}))$  [2]. Automatic differentiation allows us to easily differentiate through the least squares solution to get true derivatives of the variable projected loss.

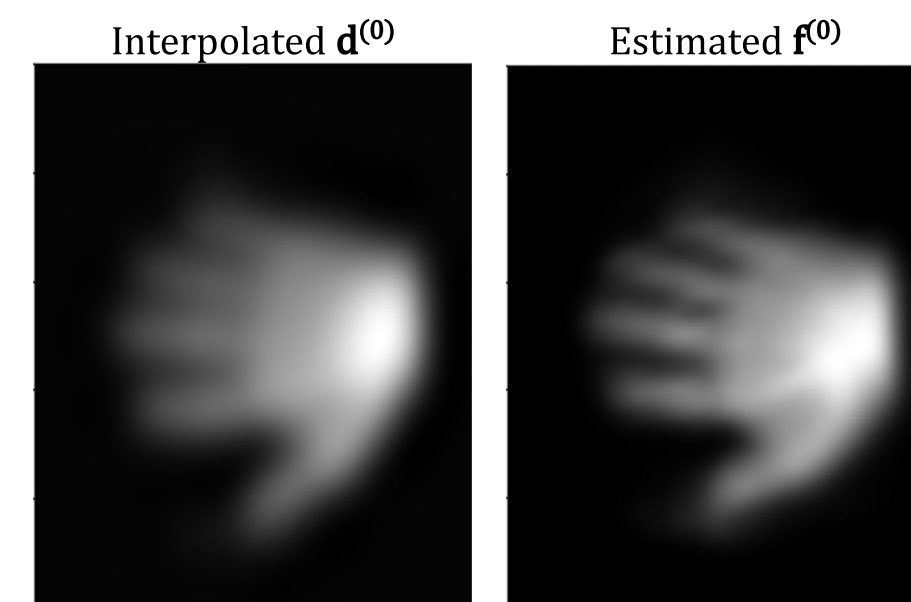
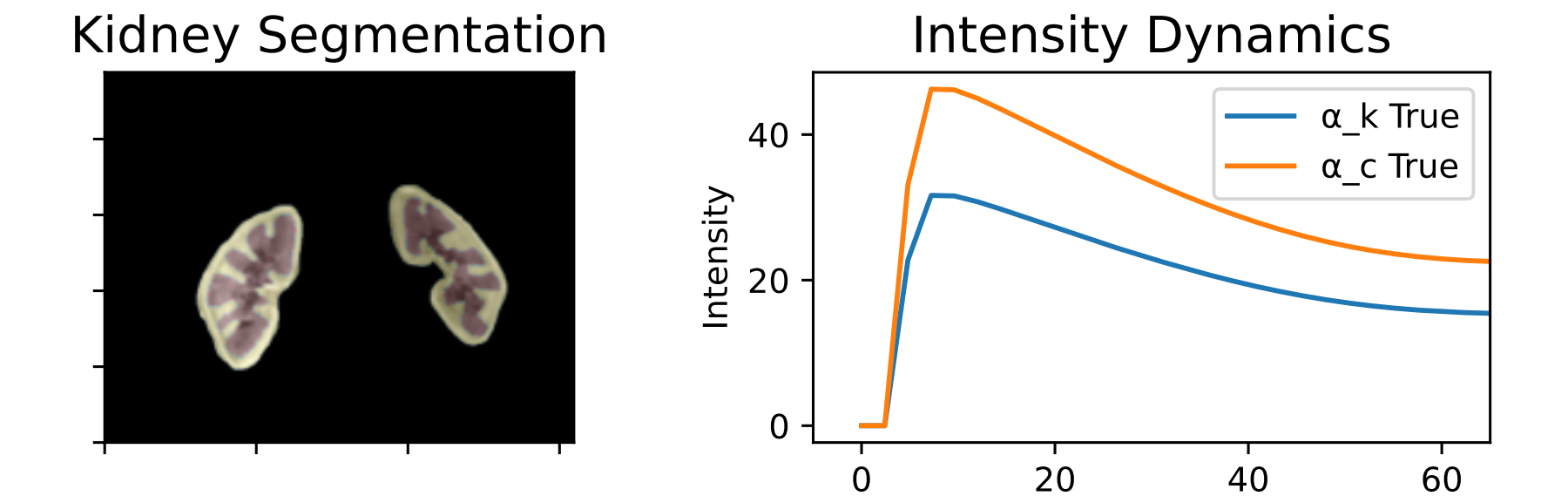


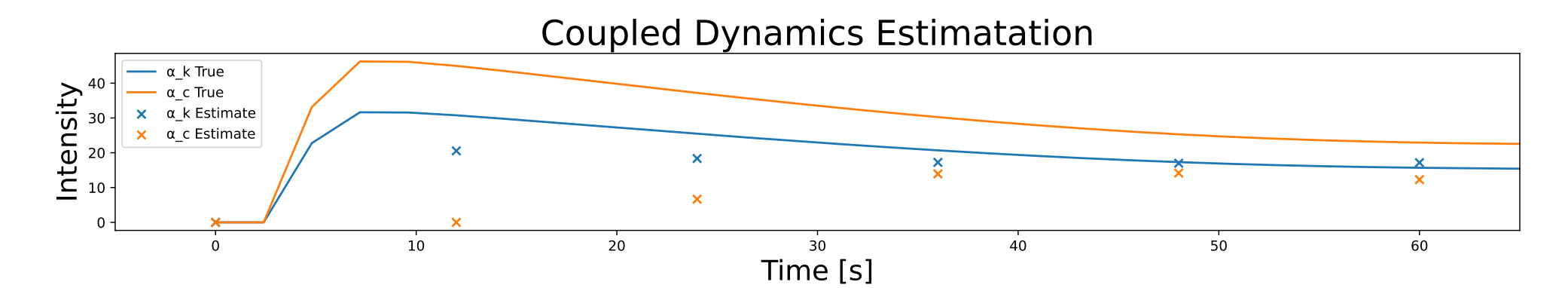
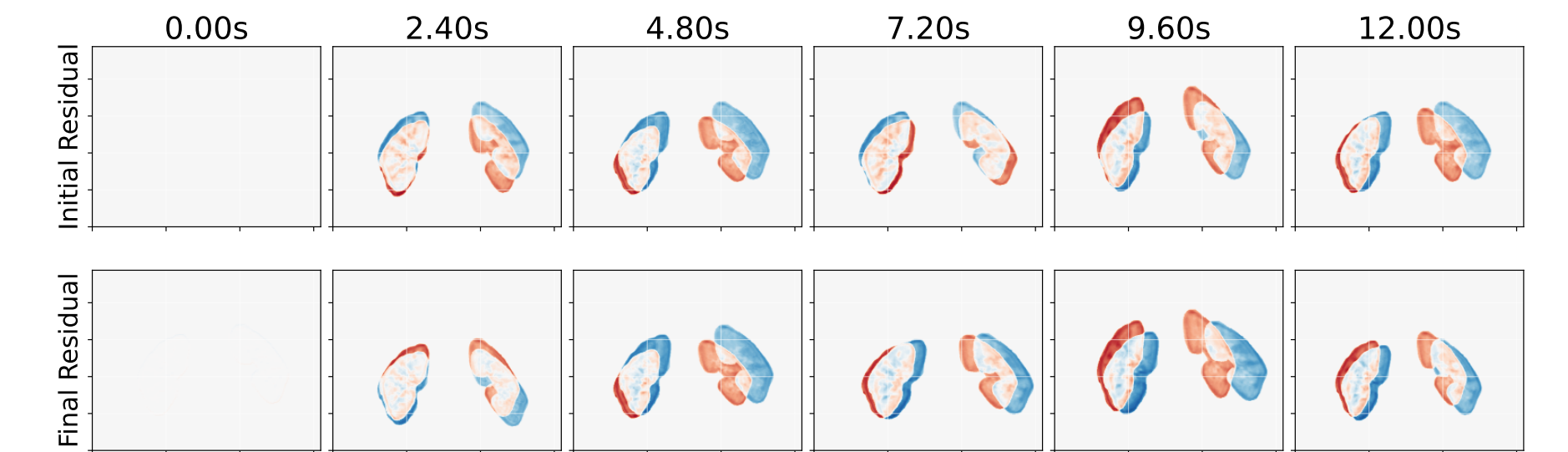
Fig. 5: super-resolution results for  $\theta = 1$ , with a sequence of 21 images

## Dynamics Estimation



We consider the problem of estimating intensity changes over pre-segmented regions. Following previous parameter map estimation approaches [5], our loss is of form

$$\ell(\mathbf{w}, \alpha = [\alpha^{(0)\top}, \dots, \alpha^{(j)\top}, \dots, \alpha^{(q)\top}]^\top) = \sum_{j=1}^q \|\mathbf{d}^{(0)} + \mathbf{M}\alpha^{(j)} - \mathbf{S}^{(j)}(\mathbf{w})\mathbf{d}^{(j)}\|^2.$$



## Conclusions

Modern machine learning packages are flexible and powerful tools for image registration. We've implemented complex image registration problems and optimization methods. Automatic differentiation significantly reduces the complexity of Hessian and Jacobian computation, and Neural ODEs provide flexible representations of diffeomorphic transformations.

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